

Notes: Malloy (JF, 2005)

The Geography of Equity Analysis

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1 Big picture

- **Question.** Does physical distance affect the quality and market value of sell-side equity research? More specifically, are analysts who are geographically close to the firms they cover more accurate and more informative than distant analysts?
- **Setting.** The paper studies U.S. equity analysts from November 1994 to December 2001. It combines I/B/E/S earnings forecasts and stock recommendations with hand-matched analyst office locations from Nelson's Directory of Investment Research, firm headquarters locations, CRSP returns, COMPUSTAT firm characteristics, SDC underwriting data, Institutional Investor analyst reputation data, and brokerage-house status measures.
- **Core challenge.** Analyst ability, analyst resources, firm difficulty, forecast timing, brokerage-house reputation, underwriter affiliation, and endogenous coverage decisions can all affect forecast accuracy. A raw comparison between nearby and distant analysts would therefore not isolate a geographic information advantage.
- **Main contribution.** The paper uses within-firm-year comparisons to ask whether local analysts perform better than other analysts covering the same firm in the same fiscal year. It then uses stock-price reactions to forecast revisions and recommendations to ask whether the market treats local analysts' signals as more informative.
- **Why it matters.** Geography is a proxy for information costs. If local analysts outperform distant analysts even in a national securities market, then soft information, local contacts, site visits, and local economic knowledge remain important despite modern communication technology.
- **Main empirical question.** Is local analyst outperformance an information effect, or is it instead driven by differences in optimism, brokerage resources, analyst specialization, analyst portfolio choices, or investment-banking conflicts of interest?
- **Main empirical results.**
 - Local analysts are significantly more accurate than distant analysts. In the full sample, the local accuracy advantage is about \$0.025 per share.
 - The effect is much larger for small, nonmetro, and remote firms. For local analysts covering small remote stocks, the estimated accuracy advantage is about \$0.141 per share.
 - Local analysts' forecast revisions move prices more than distant analysts' revisions. The incremental 3-day average excess return is about 0.095% for strong positive revisions and about -0.128% for strong negative revisions in the full sample.
 - The negative-revision effect is stronger for nonmetro firms, where the incremental local reaction is about -0.164%.
 - Local analyst outperformance is strongest for downward revisions, recent years, small firms, value firms, and firms outside major cities.
 - The usual underwriter-affiliation bias in recommendations is concentrated among **distant** affiliated analysts. Local affiliated analysts do not appear to issue significantly biased recommendations.
- **Economic takeaway.** Distance still matters in equity analysis. Local analysts seem to possess a soft-information advantage, and the geography of analysts also helps explain where investment-banking agency conflicts are most severe.

2 Data and measurement (key objects)

2.1 Main data sources

- **Forecasts and recommendations.** The paper uses I/B/E/S U.S. Detail History data for individual analyst earnings forecasts and I/B/E/S Recommendation History data for analyst stock recommendations. I/B/E/S recommendations are translated into a common 1-to-5 scale, where 1 means Strong Buy and 5 means Strong Sell.
- **Analyst locations.** Analyst names and brokerage firms are matched by hand to Nelson's Directory of Investment Research. Nelson's reports the analyst's office city, state, zip code, and brokerage office. This is important because the paper can distinguish analysts working in regional offices from analysts working at brokerage headquarters.
- **Firm locations.** Firm headquarters locations come from the Compact Disclosure database. Latitude and longitude for both analysts and firms are obtained from U.S. Census Bureau Gazetteer files.
- **Returns and firm characteristics.** Stock returns are from CRSP. Book equity, market value, employment, and related firm characteristics are from COMPUSTAT.
- **Investment-banking relationships.** IPO and SEO underwriting information comes from SDC. These data are used to identify whether an analyst's brokerage house recently served as lead underwriter for the covered firm.
- **Analyst and brokerage status.** Analyst reputation is measured using Institutional Investor All-American analyst rankings. Brokerage-house status is based on the Institutional Investor ranking of leading research firms.

2.2 Sample construction

- **Main period.** The main sample begins in November 1994 because Nelson's Directory is available starting then. The sample runs through December 2001.
- **Forecast sample.** The 1-year earnings forecast sample contains 348,571 forecasts, 4,254 analysts, and 4,344 firms.
- **Recommendation sample.** The recommendation sample contains roughly 86,000 recommendation observations, with more than 4,000 analysts and more than 4,000 firms.
- **Geographic exclusions.** Analysts and firms in Alaska and Hawaii are excluded so that extreme physical distance does not mechanically dominate the distance measure. The District of Columbia is treated as a state in the summary statistics.
- **Continental U.S. focus.** Because the paper studies U.S. analysts covering U.S. firms, the design isolates physical distance without mixing in cross-border institutional, language, accounting, and regulatory differences.

2.3 Geographic variables

- **Distance.** The paper computes the physical distance between analyst i 's office and firm j 's headquarters using latitude and longitude.

- **Local analyst.**

$$\text{LOCAL}_{ijt} = \mathbf{1}\{d_{ijt} \leq 100 \text{ km}\}.$$

A forecast is classified as local if the analyst is located within 100 km of the covered firm's headquarters.

- **Log distance.**

$$\log d_{ijt} = \log(1 + d_{ijt}),$$

where d_{ijt} is distance in kilometers. This is an alternative continuous measure of proximity.

- **Metro firms and analysts.** Metro means located in one of the 20 most populated U.S. cities.
- **Remote firms and analysts.** Remote means located more than 400 km from any of the 20 most populated cities.

Interpretation. Locality is treated as a proxy for lower information-acquisition costs. Local analysts can more easily visit firms, observe local business conditions, talk with employees or suppliers, and interact with local investors and managers.

2.4 Key analyst characteristics

- **Forecast age.** *AGE* is the log of one plus the number of days between the analyst's forecast and the firm's actual earnings announcement. Older forecasts are expected to be less accurate.
- **Firm-specific experience.** *FEXP* is the log of one plus the number of days the analyst has supplied forecasts for the firm.
- **Brokerage size.** *BSIZE* is the log of one plus the number of analysts working at the same brokerage firm in the same fiscal year.
- **All-Star.** *STAR* equals one if the analyst was listed as an Institutional Investor All-American analyst in the prior October.
- **High-status brokerage.** *HIGH* equals one if the analyst works for a top-ranked brokerage house.
- **Optimism.** *PCTOP* measures the fraction of firms followed by the analyst for which the analyst's last forecast is above actual earnings.
- **Affiliation.** *AFFIL* equals one if the analyst's brokerage house was lead underwriter for the firm's IPO in the past 5 years or for an SEO in the past 2 years.

2.5 Descriptive facts

- Only about 16.8% of forecasts are local by the 100 km definition.
- A large share of forecasts come from analysts located near New York City. In the 1-year forecast sample, about 56.5% of forecasts come from analysts within 100 km of New York.
- Metro analysts tend to work at larger brokerage houses. Remote analysts tend to work at smaller brokerage houses.
- Remote firms are smaller and have less analyst coverage than metro firms.

- The fraction of forecasts from New York-area analysts falls modestly from the first subperiod to the second subperiod, while San Francisco’s share rises.

Interpretation. The raw data already show that location is not randomly assigned. Local analysts differ from distant analysts, and remote firms differ from metro firms. This is why the paper uses within-firm-year comparisons and many analyst controls rather than simple raw differences.

3 Models and empirical specifications (all equations + interpretation)

3.1 Forecast error and relative accuracy

The absolute forecast error for analyst i ’s latest forecast of firm j ’s fiscal-year earnings is

$$AFE_{ijt} = \frac{|\widehat{EPS}_{ijt} - EPS_{jt}|}{P_{j,t-12}},$$

where $\widehat{EPS}_{ijt}$ is the analyst’s forecast, EPS_{jt} is actual earnings, and $P_{j,t-12}$ is the stock price 12 months before the beginning of the fiscal year.

The main relative-error measure is

$$DAFE_{ijt} = AFE_{ijt} - \overline{AFE}_{jt},$$

where $\overline{AFE}_{jt}$ is the mean absolute forecast error across analysts covering the same firm-year.

The proportional version is

$$PMAFE_{ijt} = \frac{DAFE_{ijt}}{\overline{AFE}_{jt}}.$$

Interpretation.

- Negative DAFE or PMAFE means the analyst is more accurate than the average analyst covering the same firm in the same year.
- Positive DAFE or PMAFE means the analyst is less accurate than the same-firm-year average.
- This relative measure directly controls for the fact that some firms and years are intrinsically harder to forecast than others.

3.2 Firm-year demeaning

For any variable X_{ijt} , define the within-firm-year demeaned variable as

$$DX_{ijt} = X_{ijt} - \overline{X}_{jt}.$$

Interpretation. The paper compares analysts covering the same firm in the same fiscal year. This removes common firm-year forecasting difficulty, such as mergers, strikes, disclosure events, or industry shocks affecting all analysts who cover that firm-year.

3.3 Main forecast-accuracy regression

The baseline Fama–MacBeth accuracy regression is

$$Y_{ijt} = \beta_1 DAGE_{ijt} + \beta_2 DFEXP_{ijt} + \beta_3 DSTAR_{ijt} + \beta_4 DHIGH_{ijt} \\ + \beta_5 DBSIZE_{ijt} + \beta_6 DPCTOP_{ijt} + \beta_7 DAFFIL_{ijt} + \beta_8 DLOCAL_{ijt} + \varepsilon_{ijt},$$

where Y_{ijt} is either $DAFE_{ijt}$ or $PMAFE_{ijt}$.

An alternative version replaces $DLOCAL$ with log distance:

$$Y_{ijt} = \beta_1 DAGE_{ijt} + \beta_2 DFEXP_{ijt} + \beta_3 DSTAR_{ijt} + \beta_4 DHIGH_{ijt} \\ + \beta_5 DBSIZE_{ijt} + \beta_6 DPCTOP_{ijt} + \beta_7 DAFFIL_{ijt} + \beta_8 DLOGD_{ijt} + \varepsilon_{ijt}.$$

Interpretation.

- The coefficient of interest is β_8 .
- With $DLOCAL$, a negative β_8 means local analysts have lower relative forecast errors and are therefore more accurate.
- With $DLOGD$, a positive β_8 means greater distance is associated with larger forecast errors.
- The regression has no intercept because all variables are demeaned within firm-year.

3.4 Relative optimism and bias

Forecast error with sign is

$$FE_{ijt} = \frac{\widehat{EPS}_{ijt} - EPS_{jt}}{P_{j,t-12}}.$$

Relative optimism is

$$OPT_{ijt} = FE_{ijt} - \overline{FE}_{jt},$$

and the proportional version is

$$POPT_{ijt} = \frac{OPT_{ijt}}{|\overline{FE}_{jt}|}.$$

Interpretation. These variables keep the sign of the forecast error. If local analysts are more accurate only because they are less optimistic, then controlling for OPT or $POPT$ should eliminate the local accuracy effect. The paper finds that it does not.

3.5 Actual earnings on forecasted earnings

The paper also compares the informativeness of local and nonlocal forecasts using regressions of actual earnings on forecasted earnings:

$$EPS_{jt} = \alpha_g + \gamma_g \widehat{EPS}_{ijt} + u_{ijt},$$

where $g \in \{\text{local}, \text{nonlocal}\}$.

Interpretation.

- Intercepts measure systematic optimism or pessimism.
- Slopes closer to one and higher R^2 values indicate more informative forecasts.
- The paper finds that local and nonlocal analysts are not significantly different in optimism, but local analysts have more informative forecasts.

3.6 Forecast-revision event study

To test whether local analysts reveal valuable information to the market, the paper studies returns around forecast revisions:

$$\text{AAR}_{j,-1,1} = \beta_0 + \beta_1 \text{LNME}_{j,-30} + \beta_2 \text{SUF}_{ij0} + \beta_3 \text{AFFIL}_i + \beta_4 \text{LOCAL}_i + \varepsilon_{ij}.$$

- $\text{AAR}_{j,-1,1}$: size-adjusted average daily return from event day -1 to day $+1$.
- $\text{LNME}_{j,-30}$: log market capitalization one month before the revision.
- SUF_{ij0} : standardized unexpected forecast revision.
- AFFIL_i : affiliation dummy.
- LOCAL_i : local analyst dummy.

The standardized unexpected forecast is

$$\text{SUF}_{ij0} = \frac{\widehat{EPS}_{ij0}^{\text{new}} - \widehat{EPS}_{ij0}^{\text{old}}}{\sigma_{j,\text{old}}},$$

where $\sigma_{j,\text{old}}$ is the cross-sectional standard deviation of outstanding forecasts for firm j .

Interpretation.

- If local revisions contain more information, the market reaction should be larger after controlling for the size of the revision.
- A positive β_4 for good-news revisions means local positive revisions move prices up more.
- A negative β_4 for bad-news revisions means local negative revisions move prices down more.

3.7 Forecast signal categories

Forecast revisions are classified by whether the new forecast is above or below both the analyst's prior forecast and the prior consensus forecast:

$$\text{SIGNAL}_{ij0} = 1 \quad \text{if} \quad \widehat{EPS}_{ij0}^{\text{new}} > \widehat{EPS}_{ij0}^{\text{old}} \quad \text{and} \quad \widehat{EPS}_{ij0}^{\text{new}} > \widehat{EPS}_{j,\text{old}}^{\text{consensus}},$$

$$\text{SIGNAL}_{ij0} = -1 \quad \text{if} \quad \widehat{EPS}_{ij0}^{\text{new}} < \widehat{EPS}_{ij0}^{\text{old}} \quad \text{and} \quad \widehat{EPS}_{ij0}^{\text{new}} < \widehat{EPS}_{j,\text{old}}^{\text{consensus}}.$$

Other revisions are treated as mixed or neutral signals.

Interpretation. The cleanest tests focus on revisions that unambiguously convey good news or bad news relative to both the analyst's own previous forecast and the market's prior consensus.

3.8 Recommendation-return regression

The recommendation event-study regression is

$$\begin{aligned} \text{AAR}_{j,-1,1} = & \beta_1 SB_{j,A} + \beta_2 B_{j,A} + \beta_3 H_{j,A} + \beta_4 SB_j \\ & + \beta_5 B_j + \beta_6 H_j + \beta_7 S_j + \varepsilon_j. \end{aligned}$$

- SB , B , H , and S indicate strong buy, buy, hold, and sell/strong sell recommendations.
- The subscript A means the recommendation is issued by an affiliated analyst.
- The coefficients β_1 , β_2 , and β_3 measure the incremental market reaction to affiliated recommendations relative to comparable unaffiliated recommendations.

The paper then repeats the same design separately for local affiliated and distant affiliated analysts.

Interpretation. If affiliated analysts are biased, investors should discount their recommendations. A negative incremental reaction to an affiliated hold recommendation means the market interprets the affiliated hold as worse news than an unaffiliated hold, consistent with the idea that affiliated analysts avoid saying “sell.”

3.9 Matched-pair recommendation bias

The recommendation code difference is

$$\Delta REC = REC^{affiliated} - REC^{unaffiliated}.$$

Because lower recommendation codes are more optimistic, $\Delta REC < 0$ means the affiliated analyst is more optimistic than the matched unaffiliated analyst.

Interpretation. The matched-pair design compares affiliated and unaffiliated recommendations for the same stock near the same SEO event, reducing concern that affiliated analysts cover a different set of firms.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard comparisons would miss

- A simple comparison of local and distant analysts could be misleading because local analysts may cover different firms, work at different brokerage houses, issue forecasts at different times, or have different experience.
- Forecast difficulty varies sharply by firm-year. A firm undergoing a merger, strike, disclosure event, or major industry shock may be hard for every analyst to forecast.
- Analysts choose which firms to cover. If good analysts self-select into covering local firms, geography may reflect endogenous coverage choices rather than an information-cost advantage.
- Stock recommendations face a separate agency problem: underwriter-affiliated analysts may have incentives to issue overly optimistic recommendations.

4.2 Main identification logic

- **Within-firm-year comparison.** The core accuracy test compares analysts covering the same firm in the same fiscal year. This removes firm-year forecast difficulty.
- **Analyst controls.** The regressions control for forecast age, firm-specific experience, brokerage size, All-Star status, brokerage-house status, analyst optimism, and underwriting affiliation.
- **Two geography measures.** The paper uses both a local dummy and log distance. The results are consistent across both measures.
- **Bias versus information tests.** The paper separates optimism from informativeness by controlling for signed forecast errors and by regressing actual earnings on forecasts separately for local and nonlocal analysts.
- **Market-reaction tests.** If local forecasts are more accurate merely because local analysts mimic others or are less biased, local revisions should not necessarily move prices more. The event-study results show that local revisions have larger price impacts, supporting an information interpretation.
- **Subsample predictions.** The local advantage should be strongest where soft information is most valuable: small firms, remote firms, nonmetro firms, and bad-news revisions. The results follow this pattern.

4.3 No formal instrument

- The paper does not rely on an instrumental-variable design.
- Its credibility comes from a sequence of comparisons and robustness checks: firm-year demeaning, rich analyst controls, alternative distance measures, analyst and firm subsamples, forecast-revision price reactions, and recommendation tests.
- The empirical strategy is therefore best interpreted as a carefully controlled observational design rather than a clean natural experiment.

4.4 Endogenous coverage concerns

- **Task complexity.** Local analysts might be more accurate because they cover fewer firms. The paper shows that controlling for the number of firms covered does not eliminate the local advantage.
- **Specialization.** Local analysts might be more accurate because they specialize in fewer industries. The paper defines expert analysts as those covering firms in only one two-digit SIC code. Expert status does not explain the local advantage.
- **Local investor demand.** Local analysts may spend more effort on local firms because local clients demand coverage. Advertising intensity provides some support for this visibility channel, but the evidence is limited because many firms have zero advertising expense.
- **New York and San Francisco concentration.** The paper checks whether results are driven by analysts in major financial centers. Dropping New York and San Francisco analysts does not overturn the main result.

4.5 How the recommendation tests identify agency effects

- The recommendation section uses underwriter affiliation around SEOs to study agency conflicts.
- The key comparison is not simply affiliated versus unaffiliated analysts, but local affiliated versus distant affiliated analysts.
- If affiliation bias is mainly an agency problem tied to large, high-status investment banks, it should be stronger among distant affiliates who are more likely to work at high-status New York brokerage houses.
- The paper finds exactly this: the bias is concentrated among distant affiliates, while local affiliates look much less biased.

5 Tables and figures

5.1 Table I: summary statistics for analysts and firms

Read:

- The forecast sample contains 348,571 1-year earnings forecasts, 4,254 analysts, and 4,344 firms.
- Only about 16.8% of forecasts are local.
- Analysts are geographically concentrated: more than half of the forecasts are issued by analysts within 100 km of New York City.
- Remote analysts work at smaller brokerage houses, and remote firms tend to be smaller and less covered.
- The table motivates the main empirical controls because local, metro, nonmetro, and remote observations differ in many observable ways.

5.2 Table II: relative forecast accuracy

Read:

- The coefficient on *DLOCAL* is negative and statistically significant in the main yearly regressions.
- The coefficient on *DLOGD* is positive in the corresponding specifications, meaning greater distance is associated with larger forecast errors.
- In the PMAFE specification, the local coefficient is about -2.77% . Multiplying this by the mean absolute forecast error of about \$0.890 gives an average local accuracy advantage of about \$0.025 per share.
- The result also appears in monthly regressions and in 2-year-ahead forecasts.
- Forecast age worsens accuracy, while brokerage resources and firm-specific experience generally improve accuracy.

Table I
Summary Statistics for U.S. Equity Analysts

This table reports summary statistics for the two main samples used in this paper. Panel A reports statistics for the 1-year earnings forecast sample, and Panel B reports statistics for the stock recommendation sample. Both samples are broken down by forecast as well as by firm characteristics. Forecasts are classified as Metro if the analyst is located in one of the 20 most populated cities, and Remote if the analyst is not located within 400 km of any of the 20 most populated cities; firms are classified analogously, but according to the location of the firm's headquarters rather than the location of the analyst. A forecast is classified as Local if the analyst is located within 100 km of the headquarters of the firm being covered. Analysts and firms located in Hawaii and Alaska are excluded from the sample, and the District of Columbia is counted as a state. For each forecast, the brokerage size (Brok. Size) is equal to the number of analysts working for the same firm in the same calendar year as the analyst producing the forecast. Analyst coverage (Coverage) is equal to the number of analysts covering the firm each fiscal year.

Panel A: 1-Year Earnings Forecast Sample ($n = 348,571$)						
Forecast Type	No. of Analysts	No. of States	Avg. Dist. to Firms (km)	Pct. Local	Pct. NYC	Mean (Median) Brok. Size
All	4254	34	1515.60	16.84	56.47	37.44 (35)
Metro	3358	13	1602.68	17.07	68.67	42.58 (41)
Nonmetro	1141	33	1179.84	15.95	0.00	17.61 (12)
Remote	462	13	1054.66	22.25	0.00	14.88 (10)
199411–199712	2645	33	1474.77	17.02	59.08	30.11 (28)
199801–200112	3849	33	1545.60	16.71	54.55	42.83 (41)
Firm Type	No. of Firms	No. of States	Avg. Dist. to Analysts (km)	Avg. Size (\$ mill.)	Avg. No. of Employees	Mean (Median) Coverage
All	4344	49	1394.69	1824.43	6947.3	5.79 (3.83)
Metro	877	14	1505.53	2290.81	8327.7	7.00 (4.60)
Nonmetro	3544	49	1373.95	1711.51	6650.0	5.49 (3.63)
Remote	966	28	1415.26	1521.59	5927.2	5.26 (3.67)
199411–199712	2998	49	1326.44	1329.70	7523.8	5.47 (3.75)
199801–200112	3610	48	1419.47	2586.19	7902.2	6.01 (4)
Panel B: Stock Recommendation Sample ($n = 87,001$)						
Recommendation Type	No. of Analysts	No. of States	Avg. Dist. to Firms (km)	Pct. Local	Pct. NYC	Mean (Median) Brok. Size
All	4054	34	1509.76	16.83	49.41	40.33 (27)
Metro	3183	13	1605.68	17.36	64.00	44.96 (34)
Nonmetro	1101	33	1227.30	15.27	0.00	26.70 (13)
Remote	436	13	1168.93	21.85	0.00	23.22 (10)
199411–199712	2204	33	1447.99	17.14	49.86	28.88 (22)
199801–200112	3448	33	1545.67	16.65	49.14	46.99 (33)
Firm Type	No. of Firms	No. of States	Avg. Dist. to Analysts (km)	Avg. Size (\$ mill.)	Avg. No. of Employees	Mean (Median) Coverage
All	4271	48	1403.62	2160.74	7632.0	3.26 (2.33)
Metro	903	14	1557.33	3037.81	8875.5	3.64 (2.67)
Nonmetro	3457	48	1372.69	1931.27	7338.8	3.15 (2.33)
Remote	937	27	1405.00	1743.77	7199.6	3.07 (2.33)
199411–199712	2773	48	1321.33	1460.87	8002.4	3.04 (2)
199801–200112	3588	48	1436.47	3052.62	8868.1	3.67 (2.5)

Table II
Regressions of Relative Forecast Error on Analyst Characteristics

Pama and MacBeth (1973) cross-sectional regressions are run (yearly in Panel A, monthly in Panel B) from November 1994 to December 2001. This table reports the time-series average of the estimated coefficients from regressions of de-meaned absolute forecast error ($DAFE$) and proportional mean absolute forecast error ($PMAFE$) on a host of analyst characteristics. All variables are firm and fiscal-year mean adjusted (the D preceding each variable stands for differenced or de-meaned). For example, the variable $DAFE$ equals the difference between the absolute forecast error for analyst i for firm j in fiscal year t and the mean absolute forecast error for firm j in fiscal year t ; absolute forecast error is equal to the absolute value of an analyst's latest forecast minus actual company earnings, as a percentage of stock price 12 months prior to the beginning of the fiscal year. The variable $PMAFE$ equals $DAFE$ divided by the mean absolute forecast error. The variable $DAGE$ equals the log of 1 plus the age (in days) of analyst i 's forecast; $DBSIZE$ equals the log of 1 plus the number of analysts working for the same firm as i and supplying forecasts during the same year t ; $DSTAR$ is a dummy variable equal to 1 if the analyst was ranked as an All-Star the previous October by *Institutional Investor* magazine; $DHIGH$ is a dummy variable equal to 1 if the analyst works at a high-status firm (where status is measured by the *Institutional Investor* magazine's rankings of brokerage firms); $DFEXP$ equals the log of 1 plus the number of days that analyst i supplied a forecast for firm j ; $DPCTOP$ equals the percentage of companies followed by analyst i in year t for which analyst i 's last forecast is above actual earnings; $DAFFIL$ is a dummy variable equal to 1 if the analyst works for the same brokerage firm that served as the lead underwriter for firm j 's initial public offering within the past 5 years or for a secondary equity offering by firm j within the past 5 years; $DLOCAL$ is a dummy variable equal to 1 if the analyst is located within 100 km of firm j in fiscal year t ; and $DLOGD$ equals the log of 1 plus the distance (in kilometers) of the analyst i from firm j in fiscal year t ; each variable is de-meaned as described above. The t -statistics are in parentheses.

Panel A: Yearly Regressions									
Dep. Var.	$DAGE$	$DBSIZE$	$DSTAR$	$DHIGH$	$DFEXP$	$DPCTOP$	$DAFFIL$	$DLOCAL$	$DLOGD$
$DAFE$	0.2351 (8.02)	-0.0200 (-2.53)	-0.0120 (-0.77)	-0.0196 (-1.20)	-0.0149 (-4.65)	0.2567 (6.57)	-0.0609 (-2.80)	-0.0197 (-2.13)	
$PMAFE$	38.3629 (10.36)	-3.9581 (-3.66)	-3.8002 (-3.83)	-5.9194 (-2.55)	-1.9194 (-4.20)	56.0157 (7.92)	-3.9867 (-2.47)	-2.7652 (-2.28)	
$DAFE$	0.2350 (8.01)	-0.0201 (-2.51)	-0.0119 (-0.76)	-0.0198 (-1.21)	-0.0149 (-4.68)	0.2566 (6.56)	-0.0610 (-2.78)	0.0025 (1.39)	
$PMAFE$	38.3497 (10.23)	-3.9569 (-3.71)	-3.8217 (-3.80)	-5.8506 (-2.57)	-1.9221 (-4.26)	56.0218 (7.91)	-3.9836 (-2.46)	0.4649 (2.72)	
$PMAFE$	39.8310 (10.50)	-7.5650 (-16.32)						-3.1704 (-2.61)	
$PMAFE$	39.8150 (10.47)	-7.6174 (-16.03)						0.3285 (0.88)	

(continued)

5.3 Table III: bias versus informativeness

Read:

- Controlling for relative optimism does not eliminate the local accuracy effect.
- When signed optimism is the dependent variable, local status is not significantly related to optimism.
- Regressions of actual earnings on forecasted earnings show that local forecasts have slopes closer to one and higher R^2 values.
- The conclusion is that local analysts are not simply less optimistic; their forecasts are more informative.

5.4 Table IV: local accuracy by firm characteristics

Read:

- The local accuracy advantage is concentrated among small stocks.
- It is also stronger for nonmetro and remote firms.
- For remote firms, the $DLOCAL$ coefficient is about -0.0629 . For small remote firms, it is about -0.0803 .

Table III
Bias versus Informativeness

Panel A reports the time series average of the estimated coefficients from yearly Fama and MacBeth (1973) regressions of de-meaned absolute forecast error (*DAFE*), proportional mean absolute forecast error (*PMAFE*), and relative optimism (*OPT*, *POPT*) on a host of analyst characteristics. The dependent variables *DAFE* and *PMAFE* and the independent variables *DAGE*, *DBSIZE*, *DSTAR*, *DHIGH*, *DFEXP*, *DAFFIL*, *DLOCAL*, and *DLOGD* are defined as in Table II; all variables are firm and fiscal-year mean adjusted (the *D* preceding each variable stands for differenced or de-meaned). The variable *OPT* equals the forecast error for analyst *i*'s forecast of firm *j* for fiscal year *t* ($FE_{i,j,t}$) minus the mean forecast error for firm *j* for fiscal year *t* ($\overline{FE}_{j,t}$); forecast error is equal to the analyst's latest forecast minus actual company earnings, as a percentage of stock price 12 months prior to the beginning of the fiscal year. The variable *POPT* is simply *OPT* scaled by the absolute value of ($\overline{FE}_{j,t}$). Panel B reports the estimated coefficient, intercept, and R^2 values for Fama and MacBeth (1973) monthly cross-sectional regressions and pooled regressions of actual earnings on forecasted earnings; local analysts' forecasts and nonlocal analysts' forecasts are grouped separately. The *t*-statistics are in parentheses.

Panel A: Incorporating Relative Optimism into Table II Regressions								
Dep. Var.	<i>DAGE</i>	<i>DBSIZE</i>	<i>DSTAR</i>	<i>DHIGH</i>	<i>DFEXP</i>	<i>POPT</i>	<i>DAFFIL</i>	<i>DLOCAL</i>
<i>PMAFE</i>	38.8645	-4.6365	-4.2789	-6.2852	-1.9307	1.5073	-2.6597	-2.7440
	(10.19)	(-4.54)	(-4.28)	(-2.71)	(-4.16)	(3.22)	(-1.40)	(-2.17)
<i>DAFE</i>	0.1779	-0.01762	-0.0079	-0.0217	-0.0110	0.3648	-0.0705	-0.0153
	(7.13)	(-2.08)	(-0.67)	(-1.31)	(-4.61)	(5.90)	(-4.91)	(-1.66)
<i>POPT</i>	57.2318	-10.2860	-3.2889	0.1960	-2.1762		5.5544	-1.2317
	(8.76)	(-6.14)	(-0.75)	(0.32)	(-2.19)		(1.60)	(-0.85)
<i>OPT</i>	0.1638	-0.0148	-0.0107	0.0003	-0.0097		0.0991	-0.0095
	(10.91)	(-1.78)	(-1.02)	(0.02)	(-5.34)		(1.62)	(-1.11)

Panel B: Regressions of Actual Earnings on Forecasted Earnings						
$(E_{i,j,t} = \alpha_i + \beta_i F_{i,j,t} + \varepsilon_{i,j,t})$						
Sample	Fama-MacBeth			Pooled		
	β	α	R^2	β	α	R^2
<i>LOCAL</i> = 1	0.9904	-0.0008	0.8589	0.9940	-0.0006	0.8752
	(155.26)	(-3.20)	(81.10)	(206.07)	(-4.52)	
<i>LOCAL</i> = 0	0.9754	-0.0008	0.8215	0.9777	-0.0006	0.8342
	(206.07)	(-4.36)	(76.34)	(631.68)	(-7.71)	
<i>DIFF</i> = (1 - 0)	0.0150	-0.0000	0.0374			
	(1.95)	(-0.06)	(2.48)			

- The implied accuracy advantage is about \$0.054 for remote firms and about \$0.141 for small remote firms.
- The effect is stronger in the later sample period, suggesting that better communication technology did not remove the value of local information.

5.5 Table V: local accuracy by analyst characteristics and signal type

Table V
Local Regression Coefficients Sorted by Analyst Characteristics
Fama and MacBeth (1973) cross-sectional regressions are run each year from 1994 to 2001 for subsamples of Table II. This table reports only the coefficient estimates and associated *t*-statistics (in parentheses) for the *DLOCAL* variable in the Table II regression of de-meaned absolute forecast error (*DAFE*) on eight analyst characteristics. *SIGNAL* = 1 if the analyst revises his forecast for firm *j* above his prior forecast for firm *j* and above the prior consensus forecast for firm *j*; *SIGNAL* = -1 if the analyst revises his forecast for firm *j* below his prior forecast for firm *j* and below the prior consensus forecast for firm *j*. All other forecasts are classified as *SIGNAL* = 0. The *XNYC* column excludes all forecasts issued by analysts located within 100 km of New York City; the *STAR* column includes only those forecasts issued by All-Star analysts; the *NON* column excludes those forecasts issued by All-Star analysts; the *HIGH* column includes only those forecasts issued by analysts working at a high-status brokerage house; and the *LOW* column excludes those forecasts issued by analysts at high-status brokerage houses. Low-coverage analysts are those who fall below the 33rd percentile of number of stocks covered by an analyst in a given year; High-coverage are those above the 66th percentile. *LOCMED* is a dummy variable equal to 1 if the share of local stocks in an analyst's portfolio is above the median share of local stocks over all analysts' portfolios. The *t*-statistics and the number of observations (*n*) are in parentheses.

Panel A: By Forecast Signal							
Sample	Dep. Var.	<i>ALL</i>	<i>XNYC</i>	<i>STAR</i>	<i>NON</i>	<i>HIGH</i>	<i>LOW</i>
All forecasts (<i>n</i> = 96,538)	<i>DAFE</i>	-0.0197 (-2.13)	-0.0250 (-2.70)	-0.0462 (-1.64)	-0.0116 (-1.12)	-0.0714 (-1.17)	-0.0192 (-2.07)
<i>SIGNAL</i> = 1 revisions (<i>n</i> = 35,910)	<i>DAFE</i>	-0.0207 (-1.15)	-0.0192 (-0.91)	0.0333 (1.81)	-0.0170 (-0.92)	-0.0431 (-1.39)	-0.0285 (-1.45)
<i>SIGNAL</i> = -1 revisions (<i>n</i> = 25,251)	<i>DAFE</i>	-0.0601 (-3.31)	-0.0614 (-2.90)	-0.1479 (-1.53)	-0.0489 (-2.42)	-0.1603 (-0.64)	-0.0429 (-3.23)
<i>SIGNAL</i> = 0 forecasts (<i>n</i> = 37,344)	<i>DAFE</i>	-0.0081 (-0.89)	-0.0318 (-3.31)	-0.1742 (-2.07)	0.0028 (0.31)	-0.0950 (-1.76)	-0.0077 (-0.94)

Table IV
Local Regression Coefficients Sorted by Firm Characteristics
Fama and MacBeth (1973) cross-sectional regressions are run each year from 1994 to 2001 for subsamples of Table II. This table reports only the coefficient estimates and associated *t*-statistics (in parentheses) for the *DLOCAL* variable in the Table II regression of de-meaned absolute forecast error (*DAFE*) on eight analyst characteristics. Stocks are classified as Metro if the firm's headquarters is located in one of the 20 most populated cities, and Remote if the firm's headquarters is not located within 400 km of any of the 20 most populated cities. The abbreviation Earn. ann. refers to quarterly earnings announcement dates. *SMALL* stocks are those below the median market capitalization for all stocks in the sample, and *LARGE* stocks are those above the median market capitalization. *GROWTH* and *VALUE* stocks are sorted similarly but by book-to-market ratio; *LOSERS* and *WINNERS* are sorted similarly but by cumulative returns over the past year. The *t*-statistics and the number of observations (*n*) are in parentheses.

Sample	Dep. Var.	<i>ALL</i>	<i>SMALL</i>	<i>LARGE</i>	<i>GROWTH</i>	<i>VALUE</i>	<i>LOSERS</i>	<i>WINNERS</i>
All stocks (<i>n</i> = 96,538)	<i>DAFE</i>	-0.0197 (-2.13)	-0.0491 (-2.79)	0.0017 (0.31)	-0.0109 (-1.20)	-0.0355 (-1.74)	-0.0287 (-2.69)	-0.0150 (-1.25)
Metro stocks (<i>n</i> = 23,261)	<i>DAFE</i>	-0.0046 (-0.33)	-0.0327 (-1.38)	0.0021 (0.23)	-0.0047 (-0.30)	-0.0092 (-0.39)	-0.0020 (-0.13)	-0.0109 (-0.88)
Nonmetro stocks (<i>n</i> = 73,277)	<i>DAFE</i>	-0.0266 (-2.12)	-0.0579 (-2.44)	0.0020 (-0.35)	-0.0125 (-0.82)	-0.0457 (-2.22)	-0.0441 (-3.48)	-0.0188 (-1.03)
Remote stocks (<i>n</i> = 19,285)	<i>DAFE</i>	-0.0629 (-2.68)	-0.0803 (-3.19)	-0.0386 (-1.03)	-0.0200 (-1.03)	-0.1030 (-2.21)	-0.0528 (-2.11)	-0.0782 (-3.07)
Subperiod: 1994:11-1997:12 (<i>n</i> = 40,616)	<i>DAFE</i>	-0.0136 (-0.75)	-0.0433 (-1.22)	0.0109 (1.26)	-0.0203 (-1.37)	-0.0037 (-0.13)	-0.0205 (-1.11)	-0.0070 (-0.30)
Subperiod: 1998:01-2001:12 (<i>n</i> = 54,722)	<i>DAFE</i>	-0.0239 (-3.50)	-0.0505 (-3.65)	-0.0080 (-2.65)	-0.0008 (-0.07)	-0.0603 (-3.59)	-0.0374 (-2.77)	-0.0197 (-2.58)
Post-Reg FD: 2000:11-2001:12 (<i>n</i> = 62,612)	<i>DAFE</i>	-0.0162 (-0.32)	-0.0733 (-1.18)	0.0339 (0.88)	0.0270 (0.41)	-0.0900 (-1.87)	-0.0122 (-0.20)	-0.0153 (-0.33)
Within 5 days of earn. ann. (<i>n</i> = 25,915)	<i>DAFE</i>	-0.0263 (-0.94)	-0.0506 (-1.03)	-0.0043 (-0.56)	-0.0060 (-0.47)	-0.0527 (-1.05)	0.0236 (0.80)	-0.0438 (-0.99)
Non-earn. ann. (<i>n</i> = 61,886)	<i>DAFE</i>	-0.0249 (-2.85)	-0.0627 (-4.95)	-0.0012 (-0.14)	-0.0028 (-0.29)	-0.0549 (-3.87)	-0.0381 (-3.00)	-0.0148 (-1.06)

Read:

- Removing New York analysts leaves the local accuracy advantage intact.
- The local advantage is particularly strong for unambiguously bad-news forecast revisions.
- High-coverage analysts with portfolios tilted toward local stocks show especially strong local performance.
- This pattern is consistent with local information, local network knowledge, or analyst effort being

Table VI
Individual Analyst Portfolios and the Effect of Coverage Decisions

Panel A reports summary statistics broken down by analyst type. Analysts are classified each year as Metro and Remote as described in Table I. An analyst is classified as Expert in year t if the analyst covers only stocks falling within one two-digit SIC code during year t . An analyst is considered Local to firm j in year t if the analyst is located within 100 km of the headquarters of firm j in year t . No. of Obs. equals the number of unique analyst-firm-year observations. No. of Firms Cov. equals the average number of firms covered each year by each analyst type, and No. of Local Firms Cov. equals the average number of local firms covered each year by each analyst type. Pct. Expert equals the percentage of expert analysts across analyst types each year, and No. of SIC Codes Cov. equals the average number of two-digit SIC codes covered by each analyst type each year. Panel B reports only the coefficient estimates and associated t -statistics (in parentheses) for the *DLOCAL* variable in the Table II regression of de-meaned absolute forecast error (*DAFE*) on various analyst characteristics. The third column adds the variable *DTASK* to this regression, the fourth column adds the variable *DSPEC* instead, and the fifth column adds both *DTASK* and *DSPEC* to the regression. *DSPEC* equals the log of 1 plus the number of firms for which analyst i supplied a forecast in year t , and *DPSEC* is a dummy variable equal to 1 if the analyst is an Expert as defined above; both variables are firm and fiscal-year mean adjusted. The sixth, seventh, and eighth columns of Panel B report estimates of the *DLOCAL* variable from the Table II regression, this time for subsamples of firms broken down by advertising intensity (ADS/S). *LOW* (ADS/S) firms are those below the median level of advertising intensity for firms in the sample, and *HIGH* (ADS/S) are those above the median level. The t -statistics and the number of observations (n) are in parentheses.

Panel A: Summary Statistics for Individual Analyst Portfolios						
Analyst Type	No. of Obs.	No. of Firms Cov.	No. of Local Firms Cov.	Pct. Expert Analysts	No. of SIC Codes Cov.	Avg. Dist. to Firms (km)
All analysts	113517	12.37	2.01	32.66	2.13	1480.36
Metro analysts	88440	12.34	2.05	33.21	2.03	1567.50
Remote analysts	9689	11.10	2.22	31.63	2.40	1098.63
Local analysts	19185	11.87	4.17	32.51	2.14	28.79
Expert analysts	37072	10.43	1.67	100.00	1.00	1521.07

Panel B: Task Complexity, Specialization, and Investor Demand							
Sample	Dep. Var.	w/ <i>DTASK</i>	w/ <i>DSPEC</i>	w/ <i>DTASK</i> and <i>DSPEC</i>	<i>NO</i> (ADS/S)	<i>LOW</i> (ADS/S)	<i>HIGH</i> (ADS/S)
All stocks	<i>DAFE</i>	-0.0186	-0.0202	-0.0188	-0.0265	0.0377	-0.0510
($n = 96,538$)		(-1.92)	(-2.18)	(-1.94)	(-1.73)	(1.63)	(-3.25)
Remote stocks	<i>DAFE</i>	-0.0587	-0.0612	-0.0551	-0.0658	0.0075	-0.1184
($n = 19,285$)		(-2.51)	(-2.65)	(-2.41)	(-2.58)	(0.07)	(-1.51)

most valuable when news is bad and difficult to verify from public sources.

5.6 Table VI: coverage choices, specialization, and visibility

Read:

- Local analysts do not outperform simply because they cover fewer firms.
- Expert analysts, defined as analysts covering only one two-digit SIC code, are not the source of the local effect.
- Adding task-complexity and specialization controls does not remove the local advantage.
- Local analysts perform better for high-advertising firms than low-advertising firms, which suggests that local investor demand or visibility may matter.
- However, the advertising evidence is limited because most firms in the sample have zero advertising expense.

5.7 Table VII: stock price impact of forecast revisions

Read:

- Strong positive local revisions have a larger positive price effect than comparable distant revisions.
- Strong negative local revisions have a larger negative price effect than comparable distant revisions.

Table VII
Stock Price Impact of Forecast Revisions

Fama and MacBeth (1973) cross-sectional regressions are run every month from November 1994 to December 2001. This table reports the time-series average of the estimated coefficients from monthly regressions of percentage size-adjusted average daily returns around forecast revisions on firm size, the magnitude of the revision, the affiliation status of the analyst (*AFFIL*), and a measure of geographic proximity (*LOCAL*). Revisions are classified by signal type, where *SIGNAL* = 1 if the analyst revises his forecast for firm *j* above his prior forecast for firm *j* and above the prior consensus forecast for firm *j*; *SIGNAL* = -1 if the analyst revises his forecast for firm *j* below his prior forecast for firm *j* and below the prior consensus forecast for firm *j*. Stocks are classified as Metro if the firm's headquarters is located in one of the 20 most populated cities. The variable *LOCAL_i* is a dummy variable equal to 1 if the analyst is located within 100 km of firm *j*. The variable *AFFIL_i* is a dummy variable equal to 1 if the analyst works for the same brokerage firm that served as the lead underwriter for firm *j*'s initial public offering within the past 5 years or for a secondary equity offering by firm *j* within the past 2 years. The variable *AAR_{j,t,a,b}* is the percentage size-adjusted average daily return for firm *j* from event day *a* to *b*. The variable *SUF_{i,j,0}* is the standardized unexpected forecast for analyst *i*, which is defined as the change in the forecast divided by the cross-sectional standard deviation of outstanding forecasts for firm *j*. The variable *LNME_{-j,-30}* is the log of 1 plus the market capitalization of firm *j*, 1 month prior to the revision. The *t*-statistics and the number of observations (*n*) are in parentheses.

Sample	Signal Type	β_0	β_1	β_2	β_3	β_4
Panel A: $AAR_{j,-1,1} = \beta_0 + \beta_1 LNME_{j,-30} + \beta_2 SUF_{i,j,0} + \beta_3 AFFIL_i + \beta_4 LOCAL_i + \varepsilon$						
All stocks	<i>SIGNAL</i> = 1	0.8469 (<i>n</i> = 86,127)	-0.0539 (7.38)	-0.0769 (-4.41)	0.0735 (-0.75)	0.0946 (0.68)
	<i>SIGNAL</i> = -1	-0.5643 (<i>n</i> = 63,214)	0.0492 (-2.56)	0.9781 (1.75)	-0.5550 (5.54)	-0.1281 (-3.67)
	<i>SIGNAL</i> = 1	0.8836 (<i>n</i> = 23,097)	-0.0586 (4.68)	-0.1172 (-2.59)	0.1320 (-0.43)	0.0395 (0.70)
Metro stocks	<i>SIGNAL</i> = -1	-0.7551 (<i>n</i> = 13,894)	0.0816 (-2.70)	0.5343 (2.39)	-0.4660 (-1.99)	-0.1346 (-1.61)
	<i>SIGNAL</i> = 1	0.8368 (<i>n</i> = 63,030)	-0.0538 (7.26)	-0.0494 (-4.17)	0.0403 (0.34)	0.1019 (1.95)
	<i>SIGNAL</i> = -1	-0.4893 (<i>n</i> = 49,320)	0.0345 (1.74)	0.9927 (1.08)	-0.3640 (-2.26)	-0.1638 (-2.54)
Panel B: $AAR_{j,2,64} = \beta_0 + \beta_1 LNME_{j,-30} + \beta_2 SUF_{i,j,0} + \beta_3 AFFIL_i + \beta_4 LOCAL_i + \varepsilon$						
All stocks	<i>SIGNAL</i> = 1	0.1205 (<i>n</i> = 86,127)	-0.0091 (4.88)	-0.0475 (-3.15)	-0.0102 (-2.80)	0.0020 (-0.60)
	<i>SIGNAL</i> = -1	-0.0536 (<i>n</i> = 63,214)	-0.0012 (1.74)	0.0455 (-0.35)	-0.0005 (2.53)	-0.0188 (-1.09)
	<i>SIGNAL</i> = 1	0.3888 (<i>n</i> = 34,067)	-0.0196 (3.43)	0.0354 (-1.32)	0.4245 (0.67)	0.0566 (3.47)
Subperiod: 199411–199712	<i>SIGNAL</i> = -1	-0.4389 (<i>n</i> = 27,635)	0.0320 (-3.65)	0.4648 (2.37)	-0.4584 (-2.11)	-0.0580 (-1.19)
	<i>SIGNAL</i> = 1	1.2076 (<i>n</i> = 51,867)	-0.0811 (7.23)	-0.1620 (-4.63)	-0.1897 (-0.92)	0.1256 (-1.20)
	<i>SIGNAL</i> = -1	-0.6677 (<i>n</i> = 35,382)	0.0629 (-1.75)	1.369 (1.29)	-0.6648 (-3.22)	-0.1802 (-2.15)
Post-Reg FD: 200011–200112	<i>SIGNAL</i> = 1	1.6342 (<i>n</i> = 5,506)	-0.1121 (3.42)	-0.1149 (-2.35)	-0.1653 (-0.41)	0.1361 (0.67)
	<i>SIGNAL</i> = -1	-1.8525 (<i>n</i> = 5,419)	0.1976 (-1.54)	1.4686 (1.27)	-1.0540 (-2.35)	-0.4210 (-1.93)

(continued)

Table VII—Continued

Sample	Signal Type	β_0	β_1	β_2	β_3	β_4
Panel D: $AAR_{j,-1,1} = \beta_0 + \beta_1 LNME_{j,-30} + \beta_2 SUF_{i,j,0} + \beta_3 AFFIL_i + \beta_4 LOCAL_i + \varepsilon$						
Excl. NYC forecasts	<i>SIGNAL</i> = 1	0.6949 (<i>n</i> = 37,007)	-0.0344 (4.94)	0.0322 (-2.22)	0.2245 (0.23)	0.2277 (1.10)
	<i>SIGNAL</i> = -1	-0.3176 (<i>n</i> = 26,179)	0.01411 (-2.19)	0.8140 (0.73)	-0.3294 (6.55)	-0.0779 (-1.21)
	<i>SIGNAL</i> = 1	1.0315 (<i>n</i> = 15,919)	-0.0704 (5.67)	-0.0764 (-3.29)	-0.3586 (-0.53)	0.2157 (-1.84)
All-Star forecasts	<i>SIGNAL</i> = -1	-0.6176 (<i>n</i> = 12,070)	0.0454 (-3.01)	0.9746 (1.86)	-0.5244 (6.13)	-0.0551 (-1.95)
	<i>SIGNAL</i> = 1	0.8368 (<i>n</i> = 70,208)	-0.0533 (6.37)	-0.0860 (-4.08)	0.2068 (-0.69)	0.0592 (1.64)
	<i>SIGNAL</i> = -1	-0.6211 (<i>n</i> = 51,144)	0.0616 (-2.22)	1.0595 (1.62)	-0.4102 (4.35)	-0.1740 (-2.41)
High-status forecasts	<i>SIGNAL</i> = 1	1.2245 (<i>n</i> = 26,169)	-0.0927 (9.00)	-0.0321 (-5.73)	-0.1690 (-0.35)	0.3173 (-1.20)
	<i>SIGNAL</i> = -1	-0.4255 (<i>n</i> = 20,067)	0.0248 (-2.36)	0.9827 (1.22)	-0.6068 (8.10)	-0.1443 (-2.41)
	<i>SIGNAL</i> = 1	0.7667 (<i>n</i> = 59,958)	-0.0458 (6.26)	-0.1014 (-3.61)	0.2629 (-0.80)	0.0387 (1.64)
Low-status forecasts	<i>SIGNAL</i> = -1	-0.6231 (<i>n</i> = 43,147)	0.0616 (-2.79)	0.9441 (2.14)	-0.3707 (5.37)	-0.1437 (-1.72)
	<i>SIGNAL</i> = 1	2.3208 (<i>n</i> = 11,743)	-0.2110 (1.93)	0.4549 (-1.69)	0.0887 (1.56)	0.1284 (0.81)
	<i>SIGNAL</i> = -1	-0.8234 (<i>n</i> = 6,964)	0.0804 (-2.13)	0.9164 (1.62)	-0.4191 (4.20)	-0.1393 (-1.53)

- In the full sample, the incremental local effect is about 0.095% for positive revisions and -0.128% for negative revisions.
- For nonmetro firms, the negative local-revision effect is about -0.164% .
- The market-reaction evidence strengthens the information interpretation because prices respond more to local analyst signals even after controlling for revision magnitude, firm size, and affiliation.

5.8 Table VIII: price impact of affiliated recommendations

Read:

- The paper replicates the standard affiliation-bias result: affiliated hold recommendations receive a significantly more negative market reaction than unaffiliated hold recommendations.
- Local affiliated hold recommendations do not produce the same negative incremental reaction; the local affiliated hold effect is about -0.25% and is not statistically significant.
- Distant affiliated hold recommendations have a large negative incremental reaction, about -1.74% .
- This suggests that the market discounts distant affiliated analysts more strongly than local affiliated analysts.

Table VIII
Stock Price Impact of Affiliated Analyst Recommendations

Fama and MacBeth (1973) cross-sectional regressions are run every month from November 1994 to December 2001. This table reports the time-series average of the estimated coefficients from monthly regressions of percentage size-adjusted average daily returns around stock recommendations on a series of dummy variables. In Panels A, B, and C, the last four independent variables are equal to 1 if the recommendation is a strong buy (SB), buy (B), hold (H), or sell (S). In Panel A, the first three independent variables ($SB_{i,A}$, $B_{i,A}$, and $H_{i,A}$) are equal to 1 if the recommendation is a strong buy, buy, hold, or sell, and is issued by an analyst who works for the same brokerage firm that served as the lead underwriter for the firm's most recent secondary equity offering (within the past 2 years), and equal to 0 otherwise. Thus β_1 , β_2 , and β_3 capture the mean incremental returns associated with affiliated analysts' recommendations relative to the same recommendations by nonaffiliated analysts. In Panel B, the first three independent variables ($SB_{i,A,L}$, $B_{i,A,L}$, and $H_{i,A,L}$) are equal to 1 if the recommendation is a strong buy, buy, hold, or sell, and is issued by a lead underwriter analyst who is also local (local analysts reside within 100 km of the firm's headquarters). The first three independent variables ($SB_{i,A,NE}$, $B_{i,A,NE}$, and $H_{i,A,NE}$) in Panel C are defined similarly; the subscript *NE* refers to a lead underwriter analyst who is not a local analyst. The variable $AAR_{i,t,h}$ is the percentage size-adjusted average daily return from event day *a* to *b*. The *t*-statistics and the number of observations (*n*) are in parentheses.

Dep. Var.	β_1	β_2	β_3	β_4	β_5	β_6	β_7
Panel A: Incremental Returns Associated with All Affiliates $AAR_{i,t,h} = \beta_1 SB_{i,A} + \beta_2 B_{i,A} + \beta_3 H_{i,A} + \beta_4 SB_{i,A} + \beta_5 B_{i,A} + \beta_6 H_{i,A} + \beta_7 S_{i,A} + \varepsilon$							
Announcement period: (<i>n</i> = 86,824)	$AAR_{i,t-1}$ 0.02211 (1.26)	-0.0862 (-2.18)	-1.7602 (-3.78)	0.7198 (16.11)	-0.0390 (-1.04)	-1.1553 (-14.23)	-1.0672 (-7.37)
Pre-release period: (<i>n</i> = 86,824)	$AAR_{i,t-2:t}$ -0.0741 (-1.50)	-0.1071 (-1.87)	-0.0751 (-1.17)	0.0463 (4.99)	0.0335 (2.70)	-0.0669 (-4.70)	-0.0557 (-2.06)
Panel B: Incremental Returns Associated with Local Affiliates $AAR_{i,t,h} = \beta_1 SB_{i,A,L} + \beta_2 B_{i,A,L} + \beta_3 H_{i,A,L} + \beta_4 SB_{i,A} + \beta_5 B_{i,A} + \beta_6 H_{i,A} + \beta_7 S_{i,A} + \varepsilon$							
Announcement period: (<i>n</i> = 86,824)	$AAR_{i,t-1}$ 0.0322 (1.90)	-0.4214 (-1.75)	-0.2463 (-0.78)	0.7202 (16.18)	-0.0438 (-1.16)	-1.1635 (-14.26)	-1.0786 (-7.38)
Pre-release period: (<i>n</i> = 86,824)	$AAR_{i,t-2:t}$ 0.0019 (0.04)	0.0050 (0.07)	-0.0021 (-0.69)	0.0457 (4.91)	0.0327 (2.63)	-0.0683 (-4.74)	-0.0556 (-2.07)
Panel C: Incremental Returns Associated with Distant Affiliates $AAR_{i,t,h} = \beta_1 SB_{i,A,NE} + \beta_2 B_{i,A,NE} + \beta_3 H_{i,A,NE} + \beta_4 SB_{i,A} + \beta_5 B_{i,A} + \beta_6 H_{i,A} + \beta_7 S_{i,A} + \varepsilon$							
Announcement period: (<i>n</i> = 86,824)	$AAR_{i,t-1}$ 0.0026 (0.46)	-0.6906 (-2.10)	-1.738 (-3.87)	0.7211 (16.12)	-0.0402 (-1.07)	-1.1562 (-14.27)	-1.0672 (-7.37)
Pre-release period: (<i>n</i> = 86,824)	$AAR_{i,t-2:t}$ -0.0673 (-1.16)	-0.09106 (-1.75)	-0.0676 (-0.98)	0.0464 (4.98)	0.0335 (2.70)	-0.0669 (-4.71)	-0.0557 (-2.06)

Table IX
The Affiliation Bias in Analyst Stock Recommendations

Panel A reports differences in mean recommendation codes (1 = Strong Buy, 5 = Strong Sell), where recommendations are matched in pairs using a (maximum) 60-day window between recommendation dates. The variable *AFFIL* is a dummy variable equal to 1 if the analyst works for the same brokerage firm that served as the lead underwriter for the firm's most recent secondary equity offering (SEO), and 0 otherwise; *NONAFF* = 1 whenever *AFFIL* = 0, and vice versa. The variable *LOCAFF* is a dummy variable equal to 1 if *LOCAL* = 1 and *AFFIL* = 1, and equal to 0 otherwise; the variable *DISTAFF* is a dummy variable equal to 1 if *LOCAL* = 0 and *AFFIL* = 1, and equal to 0 otherwise. In the *ALL* column, the affiliated recommendation nearest to the SEO date is chosen, while in the *ALL90*, *ALL120*, and *ALL250* columns, the affiliated recommendation must be within 90, 120, or 250 days of the SEO date, respectively. The *MET* column includes only those stocks headquartered in one of the 20 most populated cities, while the *NMET* column excludes those metropolitan stocks. Panel B presents summary statistics for the latter two samples of matched recommendations. *META* refers to analysts located in a metro area, while *METF* refers to stocks located in a metro area; all other variables are described in Table V. The *t*-statistics and the number of matched pairs (*n*) are in parentheses.

Panel A: Matched Pair Mean Differences						
	<i>ALL</i>	<i>MET</i>	<i>NMET</i>	<i>ALL90</i>	<i>ALL120</i>	<i>ALL250</i>
<i>AFFIL</i>	1.7247	1.6098	1.7500	1.5238	1.5484	1.5912
<i>NONAFF</i>	1.8947	2.0000	1.8702	1.7551	1.7742	1.8343
<i>DIFF</i>	-0.1700 (-2.45) (<i>n</i> = 247)	-0.3902 (-2.16) (<i>n</i> = 41)	-0.1202 (-1.61) (<i>n</i> = 206)	-0.2313 (-2.81) (<i>n</i> = 147)	-0.2258 (-2.78) (<i>n</i> = 155)	-0.2431 (-3.11) (<i>n</i> = 181)
<i>LOCAFF</i>	1.7105	1.5000	1.7500	1.5000	1.5217	1.6000
<i>NONAFF</i>	1.7105	1.6667	1.7188	1.5909	1.5652	1.6000
<i>DIFF</i>	0.0000 (0.00) (<i>n</i> = 38)	-0.1667 (-0.42) (<i>n</i> = 6)	0.0312 (0.18) (<i>n</i> = 32)	-0.0909 (-0.48) (<i>n</i> = 22)	-0.0435 (-0.23) (<i>n</i> = 23)	0.0000 (0.00) (<i>n</i> = 25)
<i>DISTAFF</i>	1.7395	1.6486	1.7611	1.5280	1.5530	1.5897
<i>NONAFF</i>	1.9209	2.0270	1.88944	1.7840	1.8106	1.8718
<i>DIFF</i>	-0.1814 (-2.41) (<i>n</i> = 209)	-0.3784 (-1.96) (<i>n</i> = 35)	-0.1333 (-1.64) (<i>n</i> = 174)	-0.2560 (-2.81) (<i>n</i> = 125)	-0.2576 (-2.87) (<i>n</i> = 132)	-0.2821 (-3.31) (<i>n</i> = 156)
Panel B: Matched Pair Summary Statistics						
	<i>Pct. HIGH</i>	<i>Pct. STAR</i>	<i>Pct. NYC</i>	<i>Pct. META</i>	<i>Pct. METF</i>	
<i>LOCAFF</i>	34.21	15.79	31.58	76.32	15.79	
<i>NONAFF</i>	14.29	5.26	31.58	76.32	15.79	
(<i>n</i> = 38)						
<i>DISTAFF</i>	55.35	30.23	68.84	91.63	16.74	
<i>NONAFF</i>	19.90	9.77	42.33	69.30	16.74	
(<i>n</i> = 209)						

5.9 Table IX: matched-pair recommendation bias

Read:

Table X
Why Are Recommendations by Local Affiliates Unbiased?

Panel A reports summary statistics for the entire recommendation sample. The column *ALL* includes all recommendations, the *AFFIL* column includes only affiliated recommendations, etc.; the variables *AFFIL*, *LOCAFF*, and *DISTAFF* are defined as in Table VIII. *FIRST* further limits the sample to affiliated recommendations associated with first-time SEOs, while *REPEAT* includes all other affiliated recommendations. Panel B reports differences in mean recommendation codes, as in Panel A of Table VIII. The variable *X* is associated with the analyst characteristic in the designated column. For example, *AFFIL_{X=1}* refers to affiliated recommendations where the affiliated analyst is also in category *X*; the first row of the *HIGH* column thus reports the mean recommendation code for affiliated analysts who also work at high-status firms. All affiliated recommendations are matched to nonaffiliated recommendations as described in Table VIII. The *t*-statistics and the number of matched pairs (*n*) are in parentheses.

Panel A: Recommendation Sample Breakdown						
	<i>ALL</i>	<i>AFFIL</i>	<i>AFFIL (FIRST)</i>	<i>AFFIL (REPEAT)</i>	<i>LOCAFF</i>	<i>DISTAFF</i>
<i>Pct. HIGH</i>	21.09	46.62	46.13	46.60	30.56	49.43
<i>Pct. LOCAL</i>	16.83	14.90	15.81	14.36	100.00	0.00

Panel B: Matched Pair Mean Differences								
<i>X</i>	(Nearest Affiliated Rec. to SEO)				(Affiliated Rec. Within 90 Days of SEO)			
	<i>HIGH</i>	<i>LOW</i>	<i>NYC</i>	<i>XNYC</i>	<i>HIGH</i>	<i>LOW</i>	<i>NYC</i>	<i>XNYC</i>
<i>AFFIL_{X=1}</i>	1.7333	1.7339	1.7212	1.7872	1.5190	1.5294	1.5319	1.5094
<i>NONAFF</i>	1.9852	1.8145	1.9152	1.8404	1.8481	1.6471	1.8404	1.6048
<i>DIFF</i>	-0.2519 (-2.64) (<i>n</i> = 136)	-0.0808 (-0.84) (<i>n</i> = 111)	-0.1939 (-2.29) (<i>n</i> = 152)	-0.0532 (-0.47) (<i>n</i> = 95)	-0.3291 (-2.89) (<i>n</i> = 79)	-0.1176 (-0.99) (<i>n</i> = 68)	-0.3085 (-3.04) (<i>n</i> = 94)	-0.0943 (-0.67) (<i>n</i> = 53)
<i>LOCAFF_{X=1}</i>	1.7692	1.6923	1.7500	1.6923	1.4286	1.5333	1.7500	1.3571
<i>NONAFF</i>	1.7692	1.6923	1.8333	1.6538	1.5714	1.6000	1.6250	1.5714
<i>DIFF</i>	0.0000 (0.00) (<i>n</i> = 12)	0.0000 (0.00) (<i>n</i> = 26)	-0.083 (-0.28) (<i>n</i> = 12)	-0.0385 (-0.02) (<i>n</i> = 26)	-0.1429 (-0.50) (<i>n</i> = 7)	-0.0667 (-0.26) (<i>n</i> = 15)	0.1250 (0.40) (<i>n</i> = 8)	-0.2143 (-0.89) (<i>n</i> = 14)
<i>DISTAFF_{X=1}</i>	1.7419	1.7549	1.7124	1.8406	1.5278	1.5283	1.5116	1.5641
<i>NONAFF</i>	2.0081	1.8333	1.9281	1.8986	1.8750	1.6604	1.8605	1.6154
<i>DIFF</i>	-0.2661 (-2.66) (<i>n</i> = 124)	-0.0784 (-0.73) (<i>n</i> = 85)	-0.2157 (-2.44) (<i>n</i> = 140)	-0.0580 (-0.42) (<i>n</i> = 69)	-0.3472 (-2.85) (<i>n</i> = 72)	-0.1321 (-0.97) (<i>n</i> = 53)	-0.3488 (-3.26) (<i>n</i> = 86)	-0.0513 (-0.30) (<i>n</i> = 39)
<i>X</i> = 1	1.7669		1.8229		1.6169		1.7514	
<i>X</i> = 0	1.8496		1.8192		1.8182		1.7946	
<i>DIFF</i>	-0.0827 (-1.24) (<i>n</i> = 266)		0.0040 (0.06) (<i>n</i> = 206)		-0.2012 (-2.48) (<i>n</i> = 155)		-0.0432 (-0.58) (<i>n</i> = 185)	

- In matched pairs around SEOs, affiliated analysts issue more optimistic recommendations than unaffiliated analysts.
- The overall affiliated-minus-unaffiliated difference is about -0.170 , where lower recommendation codes are more optimistic.
- Local affiliates are not significantly more optimistic than matched unaffiliated analysts.
- Distant affiliates are significantly more optimistic, with a matched-pair difference around -0.181 .
- This confirms that the affiliation bias is geographically concentrated among distant affiliates.

5.10 Table X: why local affiliates are less biased

Read:

- Distant affiliates are more likely to be high-status analysts, All-Star analysts, and New York analysts.
- Local affiliates are less connected to the high-status investment-banking environment where recommendation bias is most severe.
- High-status analysts and New York analysts show stronger recommendation optimism around SEO events.
- The economic interpretation is that affiliation bias is partly a high-status, New York-centered agency problem, not merely a mechanical consequence of underwriting affiliation.

6 Short summary

- **Method.** Compare local and distant analysts within the same firm-year using relative forecast errors, then examine market reactions to forecast revisions and recommendations.
- **Key measure.** A local analyst is within 100 km of the covered firm's headquarters. Distance is also measured continuously using log distance.
- **Identification.** Firm-year demeaning controls for common forecasting difficulty; analyst controls address experience, resources, reputation, optimism, and affiliation; event-study tests ask whether the market itself treats local signals as more valuable.
- **Main result.** Local analysts are more accurate and move prices more than distant analysts. The effect is strongest where soft information should matter most: small, remote, nonmetro firms and downward revisions.
- **Agency result.** Underwriter-affiliation bias in recommendations is concentrated among distant affiliated analysts. Local affiliated analysts are not significantly biased in the same way.

7 Conclusion

7.1 Core conclusion

- The paper shows that geography matters for U.S. equity analysis.
- Analysts located close to the firms they cover produce more accurate forecasts than distant analysts covering the same firms.
- Their forecast revisions also have larger stock-price impacts, implying that the market views local analyst information as more valuable.
- The conventional underwriter-affiliation bias in recommendations is not uniform: it is concentrated among distant affiliated analysts.

7.2 Economic intuition

- Physical proximity lowers the cost of collecting soft information.
- Local analysts can more easily visit firms, understand local conditions, build networks, and learn from local investors or employees.
- This advantage is most valuable when firms are small, remote, or weakly covered, because public information and analyst competition are thinner.
- Geography also affects agency conflicts. Analysts at distant, high-status, New York-centered investment banks are more exposed to pressures connected to investment-banking business.

7.3 Takeaway

The paper's main lesson is that information in financial markets is not fully spaceless. Even in U.S. public equity markets, physical proximity helps analysts produce better and more price-relevant

research. Geography also reveals an important nuance in analyst conflicts of interest: affiliation bias is not simply about underwriting relationships, but about where analysts are located and what kind of brokerage institutions they represent.